NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

SC2006

Software Engineering

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a detailed description of the functional and non-functional requirements of CarParkSmart. This document serves as a reference for developers, testers, and project stakeholders to understand the intended behaviour and constraints of the system.

This SRS specifies the requirements for a mobile application that assists users in finding nearby car parks with the lowest parking rates, tracking parking charges, and managing parked vehicle locations. It also defines features that support parking reminders, family-based car location sharing, and navigation assistance.

The scope of this document is limited to the software components of the CarParkSmart application. External systems such as public car park data APIs, mapping services, and mobile operating system services are considered dependencies and are not specified in detail within this document.

1.2 Data Dictionary

Term	Definition
User	An individual who uses the CarParkSmart application.
Username	A unique customizable identifier for each User
Car	A car registered by a user for parking tracking purposes.
Car Name	A user defined name for a registered car
Carpark	A designated parking facility available to users.

Carpark Name	A name of the Carpark e.g. Blk 285
Carpark Location	The geographical coordinates of a car park.
Parking Rate	The cost charged for parking at a car park.
Parking Session	A period during which a vehicle is parked in a car park.
Parking Start Time	The time when a parking session begins.
Parking End Time	The time when a parking session ends.
Parking Session Driver	Driver who created the parking session
Estimated Fee	The estimated parking cost incurred by an active parking session
Accumulated Fee	The total parking cost incurred during a parking session.
Rate Alert	A notification triggered when parking fees exceed a user-defined limit.
Rate Alert Threshold	The maximum parking fee set by the user before an alert is issued.
Parked Car Location	The stored location of the vehicle after parking.

Car Location Update	A manual update of the parked vehicle's location by the user.
Car Location Photo	An image reference to help users locate their parked vehicle.
Family Group	A group of users allowed to share parking location information.
Shared Access	Permission granted to family members to view the last parked location.
Navigation Assistance	Directions provided to guide users back to their parked vehicle.
Public Carpark API	An external data source that provides car park locations and rates.

1.3 References

Public Carpark API

- Carpark availability: https://data.gov.sg/datasets/d_ca933a644e55d34fe21f28b8052fac63/view
- Carpark rates:
- https://data.gov.sg/datasets/d_9f6056bdb6b1dfba57f063593e4f34ae/view

2. Overall Description

2.1 Product Perspective

Singapore has a car owner rate of almost 40%. Many families and companies share cars and use it as their main mode of transport. Users often face challenges locating affordable parking, tracking and remembering their parking costs, as well as finding where the car was parked after finishing their activity. Existing navigation or car park finders do not adequately address these parking needs.

2.2 Product Functions

Locate nearby car parks and identify the cheapest available option based on parking rate data obtained from public APIs.

- Allow users to confirm when they have parked and initiate a parking session.
- Track parking duration and estimate parking charges using published car park rates.
    - Notify users when parking charges exceed a user-defined rate threshold.
- Allow users to manually record and update the parked car location.
    - Store photo references to assist users in locating their parked vehicle.
- Provide navigation guidance from the user's current location to the parked car location.
- Enable family members to view the last known parked location of a shared vehicle.
- Manage parking history and display past parking records for registered vehicles.

2.3 User Classes and Characteristics

1. Primary User – Driver (General User)

- Uses the application regularly to locate nearby car parks and find the cheapest parking options.
- Can be working adult or student, with basic knowledge of technology usage
- Initiates and manages parking sessions by confirming when the vehicle is parked.
- Tracks parking rates and receives alerts when costs exceed predefined thresholds.
- Manually records and updates parked car locations.
- Uses navigation and photo references to locate the parked vehicle.
- Has basic technical proficiency with mobile applications.

Most functional requirements in this SRS are primarily designed to satisfy this user class.

2.4 Operating Environment

- The software will mainly be implemented in mobile platforms, mainly Android and iOS devices.
- For our demo, we will be focusing on creating a working prototype for Android phones.
- The application will require smartphones with standard mobile hardware components, including GPS, Internet Access, Touchscreen interface
- For features such as the fee alert, it will require being able to run as a background process

2.5 Design and Implementation Constraints

Due to limitations of a mobile device, we are unable to easily and reliably detect when a user has parked. As such, the system will require manual user input to indicate the start of a parking session.

- The application must comply with constraints imposed by the Android and iOS operating systems, including permissions for location access and background processing.
- The system is constrained by the capabilities and availability of external Public Carpark APIs, which provide parking rate and location data.

2.6 Assumptions and Dependencies

Public Carpark APIs provide accurate and reasonably up-to-date parking rate and location information.

- Users grant the necessary permissions (e.g. location access) required for the system to function correctly.
- The system will also depend on third-party map and navigation services and APIs for displaying maps and providing navigation assistance.

3. Functional Requirements

3.1 User Login

3. The system shall allow a user to log in to the application using Google OAuth 2 Authentication
    3.1. The system will redirect user back to the app after logging into Google

3.2 User Registration

3. The system shall allow a user to create an account
    3.1. The system will create a new user using Google OAuth 2 Authentication
        3.1.1. The system will redirect user back to the app after logging into Google
        3.1.2. The system will prompt the user to add a Username and profile picture during profile setup
            3.1.2.1. The Username shall be longer than 5 characters
            3.1.2.2. The Username must be unique
            3.1.2.3. Adding a profile picture shall be optional

3.3 Tabbed Navigation Buttons

3. The user should be allowed to switch between ‘Map’, ‘My Parking’ and ‘Profile’ Tabs

3.4 Main View

1. Map
    1.1. The map shall show the user’s current location in the center upon loading
    1.2. The map shall identify and display the nearby carparks available
        1.2.1. Tapping the Carpark should display the Parking Rate and Availability
        1.2.2. The nearby carparks data will be retrieved from the Public Carpark API

2. ‘Start Parking Session’ Action Button
    2.1. The button shall allow the user to create a new parking session via pop up page
        2.1.1. The page should allow users to choose which car is assigned for this parking session
        2.1.2. The page should be able to see their current parked location
            2.1.2.1. The page should allow the user to edit their location precisely if necessary
        2.1.3. The page should display the Carpark details
            2.1.3.1. The details should include Parking Rate, Carpark Name and Location taken from public carpark API
        2.1.4. The page should allow the user to set session name and description
        2.1.5. The page should be able to add relevant pictures via a dialog
            2.1.5.1. The dialog should allow users to upload pictures from gallery

2.1.5.2. The dialog should allow users to take a picture directly using their phone camera

2.1.6. The page should allow the user to set a Rate Alert Threshold

2.1.6.1. The app should trigger a Rate Alert by creating a notification when the Estimated Cost exceed the Threshold

3. My Parking Page

3.1. The page will show a list of cars registered under the user

3.1.1. Each entry will expand to show the list of Parking Sessions with that car

3.2. List of Parking Sessions

3.2.1. Each entry will show the Session Name, Carpark Name

3.2.2. Tapping the list entry shows the Parking Start Time, Parking End Time, Accumulated Fee or Estimated Cost, Session Driver, User-Uploaded Pictures and Carpark Location

3.2.2.1. Tapping the Carpark Location should redirect the user to their navigation app (eg: Google Maps) with the Carpark Location as destination

3.2.2.2. Tapping the pictures should show a popup with an enlarged picture

4. Profile Page

4.1. Personal Information

4.1.1. The system shall allow a user to view their profile information

4.1.1.1. The profile information shown shall include their Username, Google email and profile picture

4.1.2. The system shall allow a user to delete their account

4.2. Family Management

4.2.1. The system shall allow the user to create a family

4.2.1.1. Each account will only be able to create 1 family

4.2.1.2. Each family created will have a join code

4.2.2. The system shall allow the user to join a family using a join code

4.2.3. The system shall display the list of family members if the user is in a family

4.2.4. The system shall allow a user to manage family group membership

4.2.4.1. The family group membership shall have a limit 5

4.3. Car Management

4.3.1. The system shall display the list of existing cars registered by the user

4.3.1.1. The details displayed shall include Carplate, Car Name, and a preset car icon.

4.3.2. The system shall allow a user to add a new car by clicking the add button

4.3.2.1. The button will direct to the user to a pop up page

4.3.2.2. The User shall fill in the Carplate, Car Name and choose from a preset car icon

4.3.2.3. If any details are missing the system shall prompt the user

4. Nonfunctional Requirements

4.1 Performance

1. The application must have a start up time of under 3 seconds
2. The main view map data should be loaded under 5 seconds upon successful login
3. The Carpark data should be loaded under 5 seconds upon selecting the Carpark
4. The Parking Session list shall be loaded and displayed under 3 seconds upon clicking on ‘My Parking’
5. The Parking Session data shall be loaded and displayed under 3 seconds upon clicking on the entry
6. After a parking session is created, it should be recorded into the backend system under 5 seconds
7. The system shall update parking fee calculations at intervals not exceeding 1 minute during an active parking session.
8. After joining a family via a join code, the system should add the user into the family and display all members under 10 seconds

4.2 Scalability

1. The application must handle up to 500 simultaneous users without a degradation in performance
2. The system shall maintain responses time within defined performance threshold when the number of active users increase

4.3 Security

1. Users of the application must be authenticated using secure OAuth2 Protocol
2. All sensitive information such as car details and location shall be have end to end encryption
3. User data should be deleted when their account is deleted
4. The system shall restrict shared parking location access to authorized family group members only.
5. The system shall automatically log out users after a defined period of inactivity

4.4 Reliability

1. The application shall not have random crashes
2. The system shall display an appropriate notification if external API data is unavailable
3. The system shall automatically retry failed API requests up to a defined number of attempts before displaying an error message.

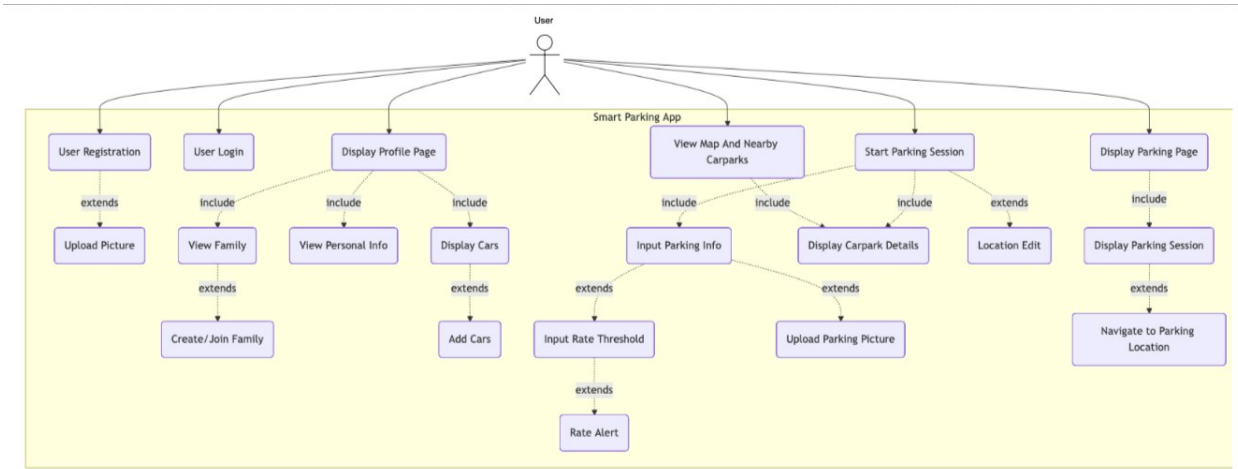
4. The system shall maintain data consistency when network connectivity is temporarily lost and synchronize data once connectivity is restored.

4.5 Usability

1. Help messages must be displayed in the local language according to the user's locale.
2. Users should navigate between pages using the application's icons with ease.

5. Use case diagram & description

5.1 Use case diagram



5.2 Use case description

Use Case ID:	001		
Use Case Name:	UserLogin		
Created By:	Bian Lingzhu	Last Updated By:	Bian Lingzhu
Date Created:	10/2/2026	Date Last Updated:	10/2/2026

Actor:	User
Description:	Allows a user to create a new account using Google OAuth 2 Authentication and complete profile setup
Preconditions:	1. User has a valid Google account

Postconditions:	2. User has logged in successfully 3. User is redirected back to the app
Priority:	High
Frequency of Use:	High
Flow of Events:	1. User selects the “Login with Google” button 2. System redirects the user to Google OAuth 2 authentication 3. User confirms their Google credentials 4. System authenticates the user 5. System redirects user back to application’s main screen
Alternative Flows:	None
Exceptions:	1. If any required information is not inputted, display an error message to notify user
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

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	5. System prompts user to enter a username and profile picture 6. User enters their username and uploads a profile picture with use case **UploadProfilePicture** 7. System validates username length and uniqueness 8. System creates an account
Alternative Flows:	None
Exceptions:	1. If any required information is not inputted, display an error message to notify user 2. If username is too short (<5 characters), system displays a custom error message and prompts re-entry. 3. If username already exists, system displays a custom error message and prompts re-entry.
Includes:	UploadProfilePicture
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

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	1. Application will use the phone camera to take a picture AF-S2: User chooses chooses upload from gallery 1. User can choose a past picture to upload from the phone's photo gallery AF-S3: User chooses to not upload a profile picture 2. A default picture will be inserted as the profile picture
Exceptions:	None
Includes:	None
Special Requirements:	Enable access to camera and/or photo gallery
Assumptions:	None
Notes and Issues:	None

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	1. Details of it are displayed with use case **DisplayCarparkDetails** 2. User can tap the back button to return to step 2
Exceptions:	None
Includes:	DisplayCarparkDetails
Special Requirements:	Location permission enabled Nearby carparks retrieved from Public Carpark API
Assumptions:	None
Notes and Issues:	None

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Date Created:	10/2/2026	Date Last Updated:	10/2/2026

Actor:	User
Description:	Allows the User to create a new parking session
Preconditions:	1. User is logged in 2. User has at least one car registered
Postconditions:	1. A new parking session is created
Priority:	High
Frequency of Use:	High
Flow of Events:	1. User selects “Start Parking Session” 2. User selects which car to assign to this parking session 3. User selects which carpark to start the session at 4. System uses the carpark location as the parked location as default, displaying as a pin on the map 5. User enters the relevant information about the parking session under use case **InputParkingInfo** 6. User confirms session 7. System creates parking session
Alternative Flows:	AF-S5: User wants to edit their location to be more precise 1. User manually edits location with the **LocationEdit** use case 2. Continue to step 4 AF-S6: User wants to view the carpark’s specific details before choosing it 1. System displays the carpark’s details with use case **DisplayCarparkDetails** 2. User can go back to step 2
Exceptions:	None
Includes:	LocationEdit DisplayCarparkDetails InputParkingInfo
Special Requirements:	Location permission enabled
Assumptions:	None
Notes and Issues:	None

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Use Case Name:	LocationEdit		
Created By:	Bian Lingzhu	Last Updated By:	Bian Lingzhu
Date Created:	10/2/2026	Date Last Updated:	10/2/2026

Actor:	User
Description:	Allows the User to edit their parked location
Preconditions:	1. User is logged in 2. User is creating a parking session
Postconditions:	1. Location of the parking spot has successfully been changed
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. User selects the “edit location” button 2. User drags the pin icon around the map to show a more precise location 3. User confirms the final location displayed on the map
Alternative Flows:	None
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	User is still parking at the same carpark, and parking rates will not change with this edit
Notes and Issues:	None

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Frequency of Use:	High
Flow of Events:	1. User inputs session name 2. User inputs session description 3. User adds relevant parking pictures with use case **UploadParkingPicture** 4. User sets a rate alert threshold with use case **InputRateThreshold** 5. User confirms the information
Alternative Flows:	None
Exceptions:	None
Includes:	InputRateThreshold UploadParkingPicture
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

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	2. **RateAlert** use case will not be triggered during this parking session
Exceptions:	None
Includes:	RateAlert
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

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Use Case Name:	UploadParkingPicture		
Created By:	Bian Lingzhu	Last Updated By:	Bian Lingzhu
Date Created:	10/2/2026	Date Last Updated:	10/2/2026

Actor:	User
Description:	Allows user to upload a picture of where they parked their car
Preconditions:	1. User is logged in 2. User is creating a parking session
Postconditions:	1. Picture is successfully uploaded 2. User can continue with creating their parking session
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. System prompts user to upload a picture 2. User can choose between take a picture from camera or upload from gallery 3. User presses confirm 4. User returns to registering their account
Alternative Flows:	AF-S8: User chooses take a picture from camera 1. Application will use the phone camera to take a picture AF-S9: User chooses chooses upload from gallery 1. User can choose a past picture to upload from the phone's photo gallery AF-S10: User chooses to not upload a parking picture 1. No picture will be displayed and operations continue as normal
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

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Use Case Name:	DisplayParkingPage		
Created By:	Bian Lingzhu	Last Updated By:	Bian Lingzhu
Date Created:	10/2/2026	Date Last Updated:	10/2/2026

Actor:	User
Description:	Allows User to view all their parking sessions
Preconditions:	1. User is logged in
Postconditions:	1. All parking sessions are displayed
Priority:	High
Frequency of Use:	High
Flow of Events:	1. User navigates to My Parking tab with navigation buttons 2. System displays a list of their registered cars in the page 3. Each car can expand to show their recent parking sessions, showing session name and carpark name 4. User can interact with each parking session with **DisplayParkingSession** use case
Alternative Flows:	None
Exceptions:	None
Includes:	DisplayParkingSession
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

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Frequency of Use:	High
Flow of Events:	1. User taps into a parking session 2. Each entry will further show the Parking Start Time, Parking End Time, Accumulated Fee or Estimated Cost, Session Driver, User-Uploaded Pictures and Carpark Location
Alternative Flows:	AF-S11: User taps a Carpark Location button 1. Will trigger the **NavigateToParkingLocation** use case AF-S12: User taps the picture of the parking session 1. Will show a popup with an enlarged picture
Exceptions:	None
Includes:	NavigateToParkingLocation
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

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Special Requirements:	None
Assumptions:	If user has used manual location adjustment, the app still redirects user based on carpark location, not the newly adjusted location
Notes and Issues:	None

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Exceptions:	None
Includes:	ViewPersonalInfo DisplayCars ViewFamily
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

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Use Case ID:	017		
Use Case Name:	Create/JoinFamily		
Created By:	Bian Lingzhu	Last Updated By:	Bian Lingzhu
Date Created:	10/2/2026	Date Last Updated:	10/2/2026

Actor:	User
Description:	Allows User to create a new family or join an existing family
Preconditions:	1. User is logged in 2. User is displaying their profile page 3. User is not in a family
Postconditions:	1. Able to create a new family or join an existing family successfully
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. User taps on “Family” button 2. Page displays a “create family” button and “join family” button 3. If user chooses to create a new family, they automatically join the new family they have created, and a join code is generated 4. If user chooses to join a new family, they will be prompted to enter a valid join code to join that family
Alternative Flows:	AF-S18: If user chooses to create a new family 1. Automatically join the new family they have created 2. A join code is generated AF-S19: If user chooses to join a new family 1. User will be prompted to enter a valid join code to join that family
Exceptions:	1. If user enters invalid join code, system will notify the user with a custom error message and ask for a new join code 2. If family already has 5 people, system will notify the user with a custom error message and ask for a new join code
Includes:	None
Special Requirements:	None
Assumptions:	None

Notes and Issues:	None

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Actor:	User
Description:	Allows user to add a new car
Preconditions:	1. User is logged in 2. User is viewing their cars
Postconditions:	1. All cars are displayed in a list
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. User selects the “add car” button 2. System directs user to a pop up page 3. User inputs the Carplate, Car Name and choose from a preset car icon
Alternative Flows:	None
Exceptions:	1. If any details are missing, system shall prompt the user with an error message
Includes:	AddCar
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

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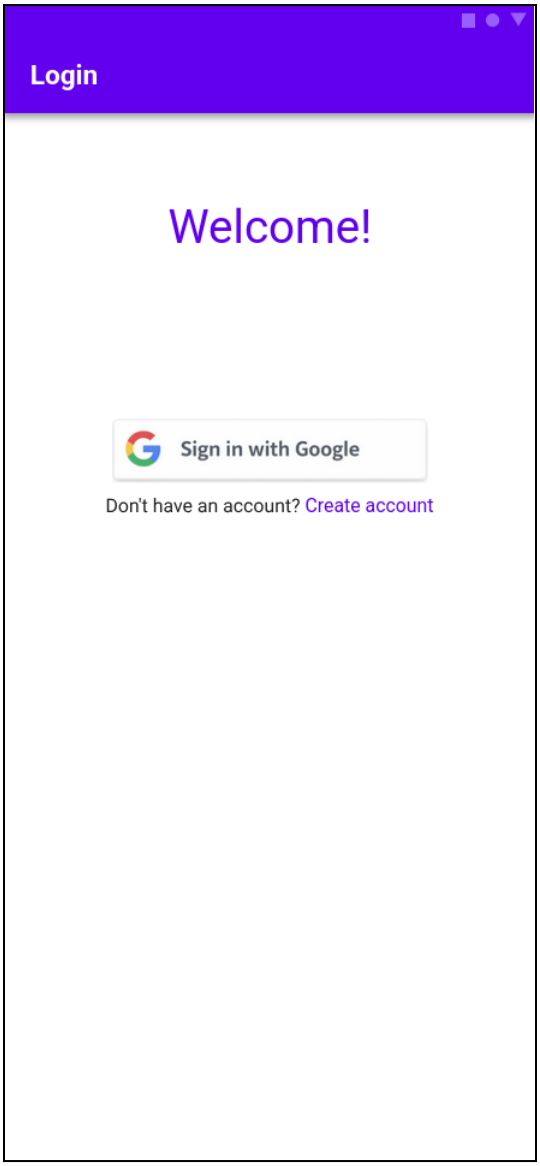
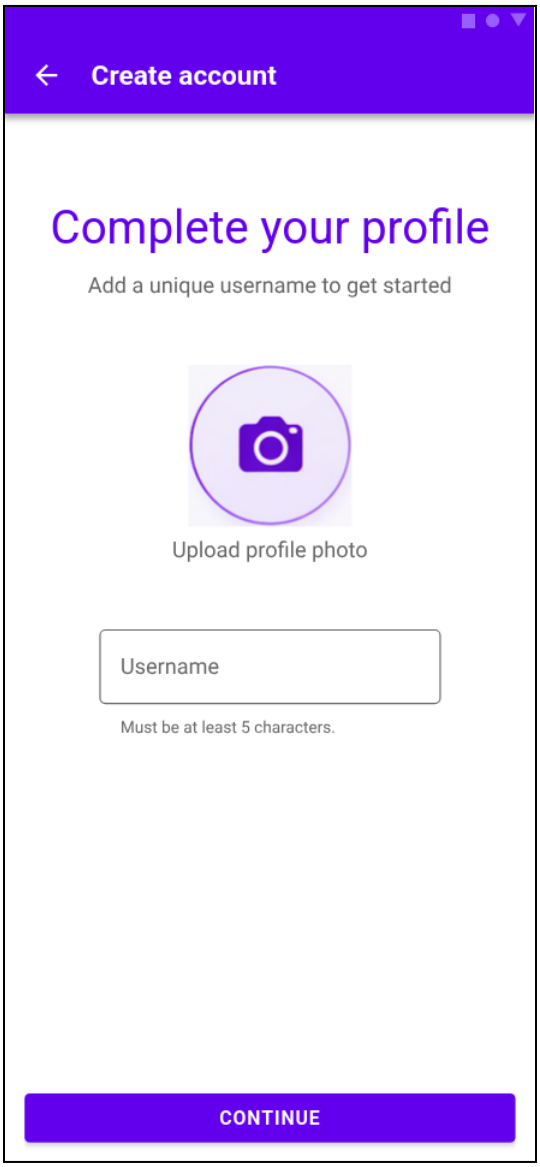
Exceptions:	1. If username already exists, system will display an error message
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

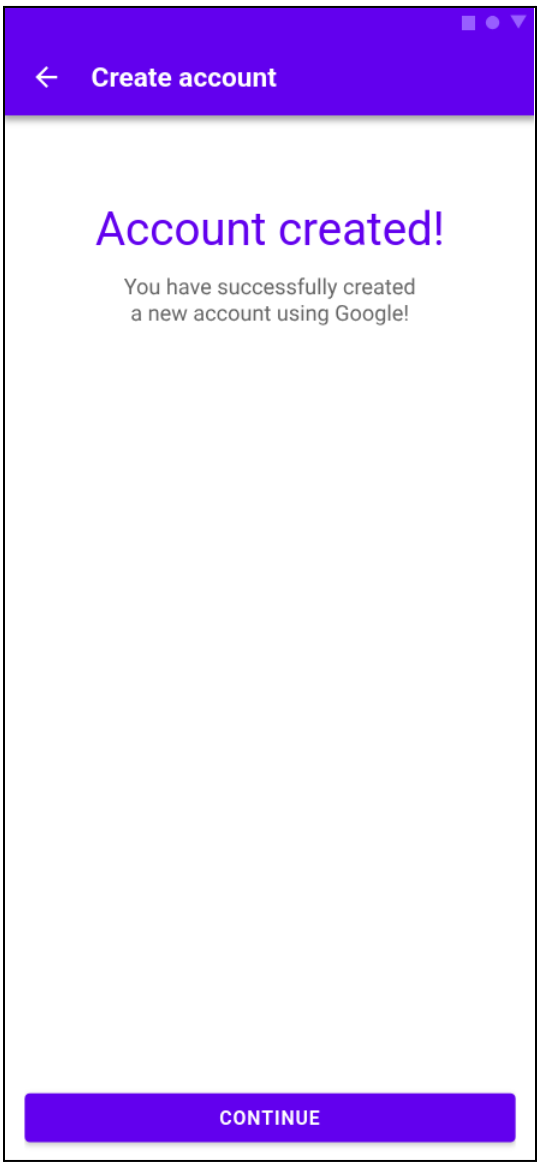
6. UI Mockup

6.1 User login



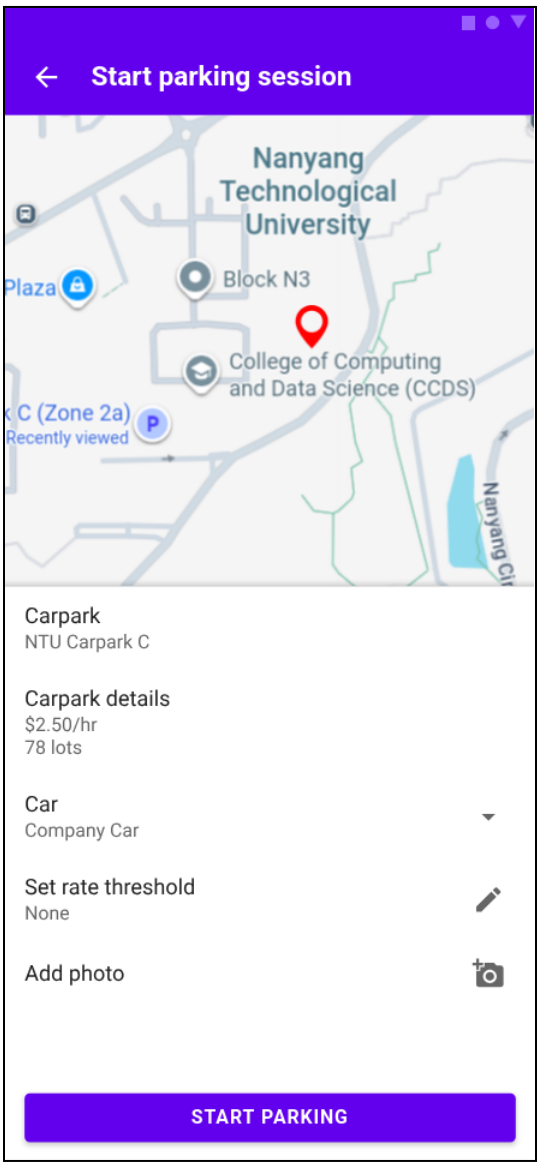
6.2 User registration



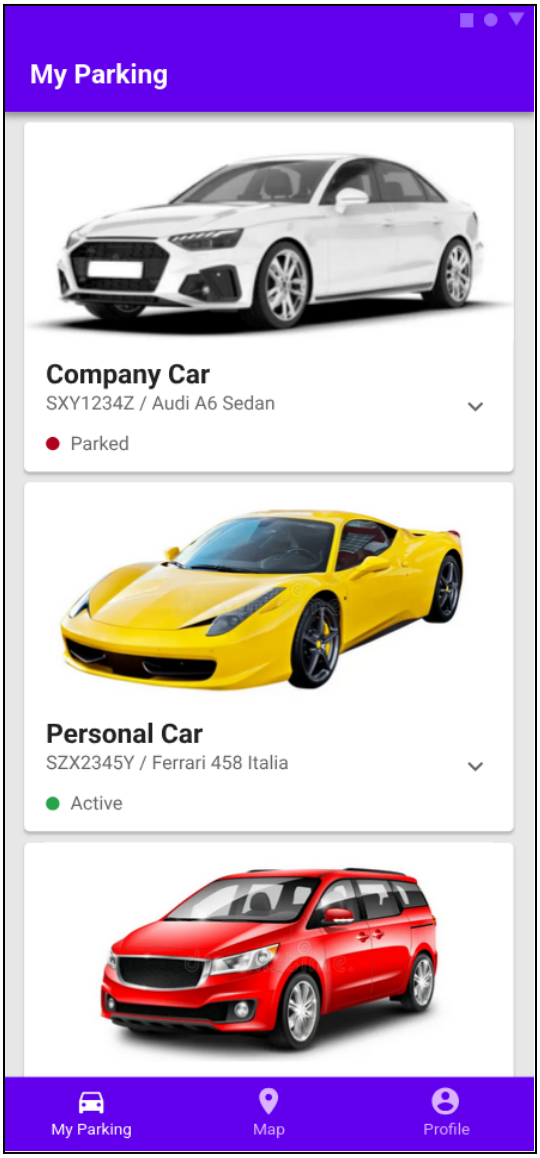


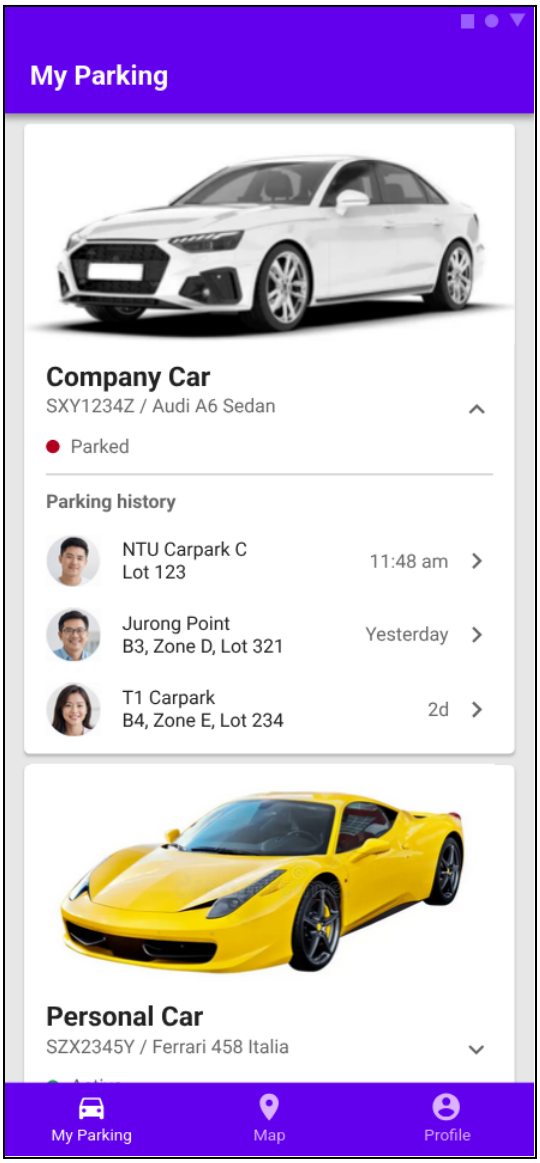
6.3 Map

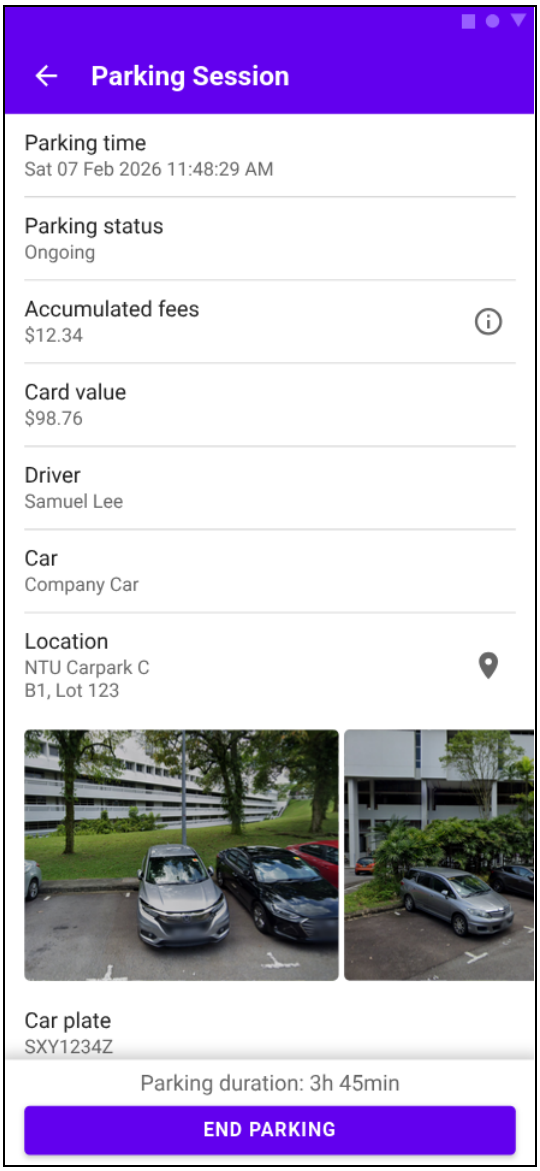
Map
Carpark B
$2.00/hr
56 lots
Carpark C
$2.50/hr
78 lots
Nanyang
Technological
University
Block N3
College of Computing
and Data Science (CCDS)
Carpark A
$1.50/hr
34 lots
The Hive
(Learning Hub
South - LHS)
PARK NOW
My Parking
Map
Profile

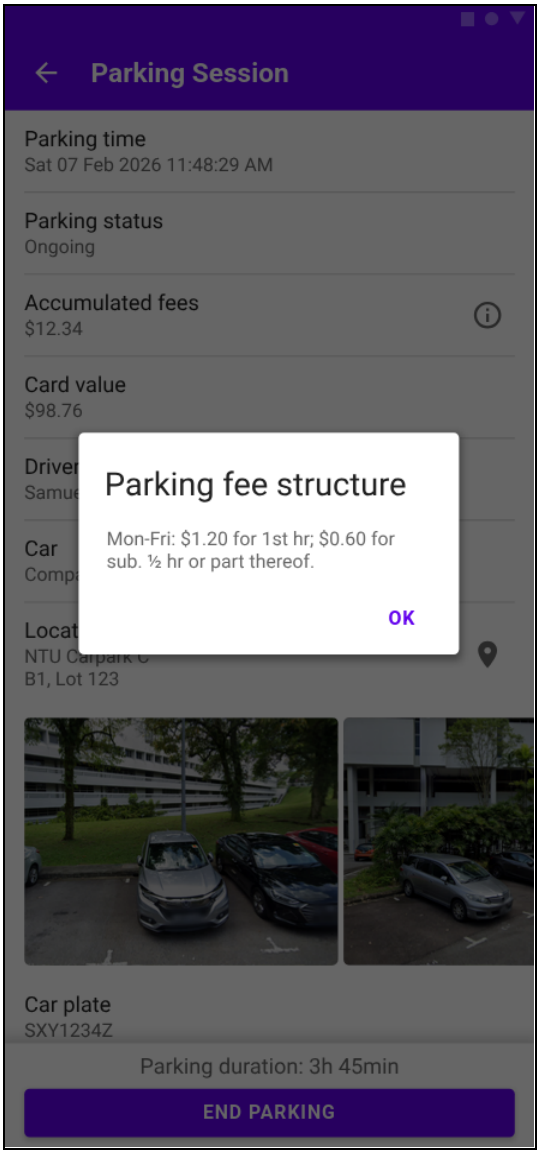

Start parking session
Nanyang
Technological
University
Block N3
College of Computing
and Data Science (CCDS)
Carpark
NTU Carpark C
Carpark details
$2.50/hr
78 lots
Car
Company Car
Set rate threshold
None
Add photo
START PARKING

6.4 My Parking

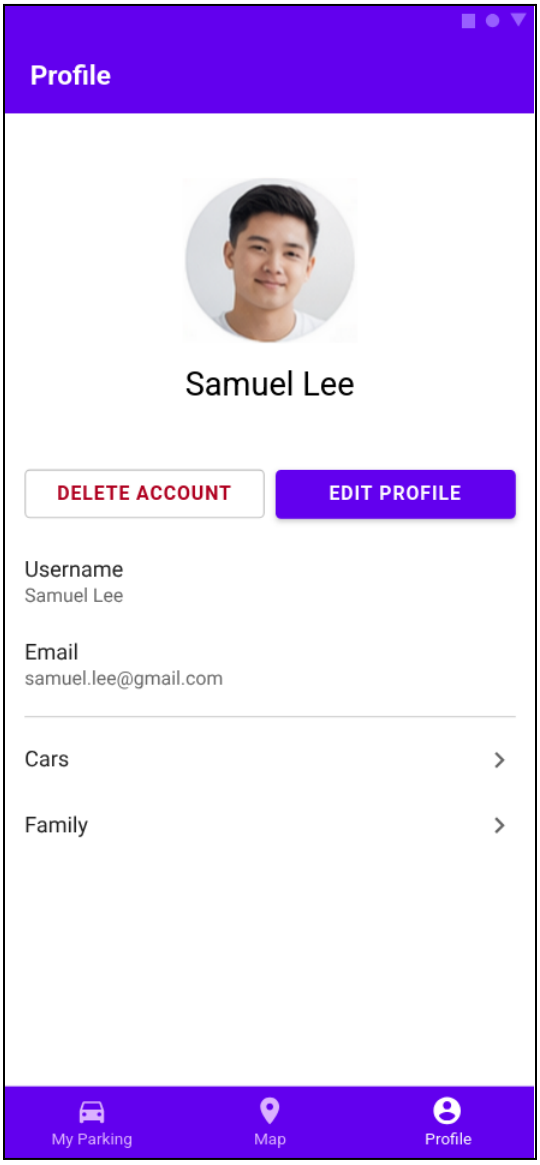


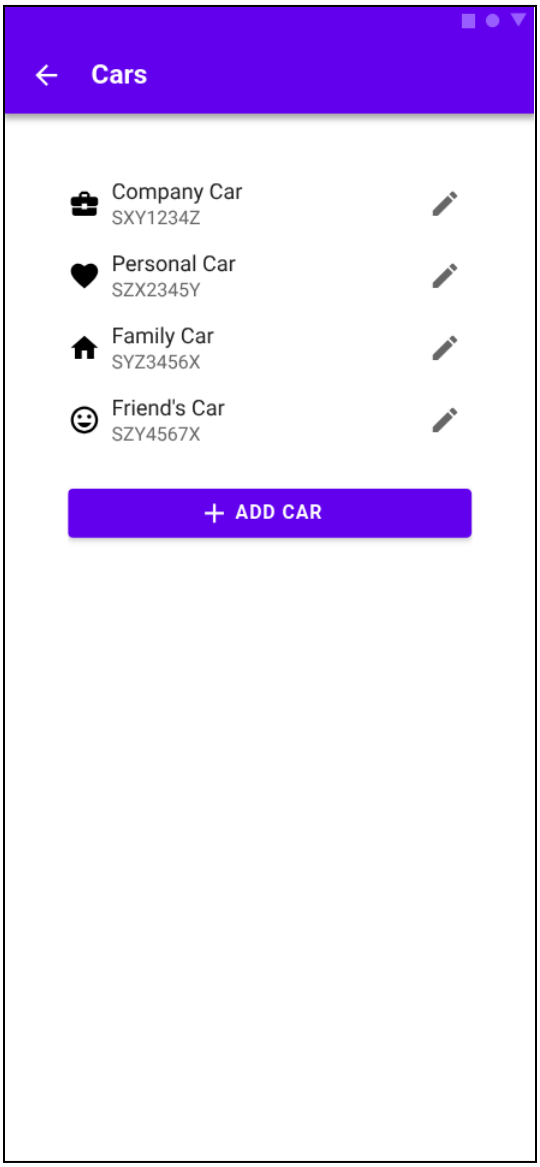


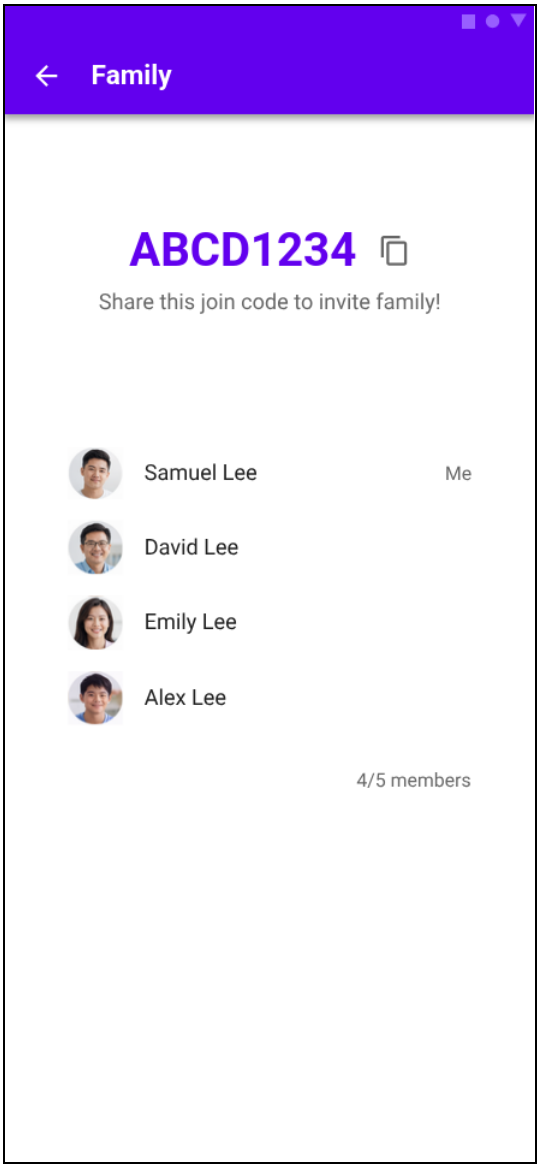

Parking Session
Parking time
Sat 07 Feb 2026 11:48:29 AM
Parking status
Ongoing
Accumulated fees
$12.34
Card value
$98.76
Driver
Samuel Lee
Car
Company Car
Location
NTU Carpark C
B1, Lot 123
Car plate
SXY1234Z
Parking duration: 3h 45min
END PARKING


Parking Session
Parking time
Sat 07 Feb 2026 11:48:29 AM
Parking status
Ongoing
Accumulated fees
$12.34
Card value
$98.76
Parking fee structure
Mon-Fri: $1.20 for 1st hr; $0.60 for sub. ½ hr or part thereof.
OK
B1, Lot 123
Car plate
SXY1234Z
Parking duration: 3h 45min
END PARKING

6.5 Profile


Profile
Samuel Lee
DELETE ACCOUNT
EDIT PROFILE
Username
Samuel Lee
Email
samuel.lee@gmail.com
Cars
Family
My Parking
Map
Profile


Cars
Company Car
SXY1234Z
Personal Car
SZX2345Y
Family Car
SYZ3456X
Friend's Car
SZY4567X
+ ADD CAR


Family
ABCD1234
Share this join code to invite family!
Samuel Lee
Me
David Lee
Emily Lee
Alex Lee
4/5 members